

Olist E-Commerce Dataset

Customer Clustering and Product Analysis Report

Prepared for Data Analysis Internship Submission

This report presents an end-to-end customer clustering and product performance analysis using the full available Olist relational dataset: customers, orders, order items, products, sellers, payments, reviews, geolocation, and product category translations. The analysis follows an industry-style workflow: data profiling, cleaning, analytical dataset construction, feature engineering, customer segmentation, product/category performance analysis, and business recommendations.

Executive Summary

Delivered orders analysed	96,478
Unique customers clustered	93,358
Products analysed	32,216
Product categories analysed	74
Total payment value analysed	15,422,605
Total item revenue analysed	13,221,498
Average review score	4.16/5
Average delivery duration	12.6 days

- Revenue is concentrated geographically: SP is the largest customer state by revenue with 37.4% of delivered-order payment value.
- The leading category is health_beauty, contributing 9.3% of product revenue in delivered orders.
- Customer behavior is highly skewed toward one-time purchase patterns, so customer_unique_id was used instead of customer_id to avoid treating repeat buyers as unrelated customers.
- Customer clusters show different action paths: retention for high-value multi-order customers, reactivation for dormant buyers, and service recovery for low-satisfaction customers.
- Product portfolio performance is dominated by a small set of revenue-driving categories and products, while a large long tail requires careful inventory and catalog governance.

1. Scope, Objective and Analytical Approach

Objective: identify actionable customer segments and product portfolio patterns that can support marketing, retention, merchandising, logistics, and customer experience decisions.

Analytical grain: customer analysis is performed at customer_unique_id level; product analysis is performed at product_id and category level; order analysis is restricted mainly to delivered orders for reliable sales, delivery, review, and payment interpretation.

Why delivered orders were used: cancelled and unavailable orders do not represent completed demand and can distort revenue, delivery time, review, and product performance metrics. They are still kept in the cleaned enriched order file for auditability.

Table 1. Raw Dataset Profiling Summary

Dataset	Rows	Columns	Duplicate Rows	Missing Cells	Missing Pct
customers	99,441	5	0	0	0.0
orders	99,441	8	0	4,908	0.6
order_items	112,650	7	0	0	0.0
payments	103,886	5	0	0	0.0
reviews	99,224	7	0	145,903	21.0
products	32,951	9	0	2,448	0.8
sellers	3,095	4	0	0	0.0
category_translation	71	2	0	0	0.0
geolocation	1,000,163	5	261,831	0	0.0

2. Data Cleaning and Preparation

- Removed exact duplicate records where applicable and de-duplicated master tables using primary keys such as customer_id, order_id, product_id, and seller_id.
- Converted purchase, approval, carrier, delivery, estimated delivery, review, and shipping-limit columns into datetime fields.
- Translated product category names from Portuguese to English using the category translation table; missing translations/categories were retained as unknown instead of being dropped.
- Aggregated payment records to order level because one order can have multiple payment rows.
- Aggregated review records to order level because some orders can contain duplicate or multiple review records.
- Aggregated order item rows to order level for customer clustering and retained line-level order-item records for product analysis.
- Aggregated geolocation at zip-code-prefix level using median latitude/longitude to avoid many-to-many duplication in the customer/order base.
- Created delivery, lateness, freight ratio, order value, category diversity, and review satisfaction features.

3. Customer Clustering Analysis

The clustering model uses behavioral and operational features: recency, frequency, monetary value, average order value, number of items, category diversity, review score, delivery duration, late delivery rate, installments, freight-to-sales ratio, and credit-card share. Numeric outliers were winsorized at the 1st and 99th percentiles, skewed variables were log-transformed, and all features were standardized before K-Means clustering.

Model selection note: K values from 3 to 7 were tested using silhouette score and business interpretability. The selected solution uses 5 clusters because it produced distinct, explainable commercial segments. Inertia values on a 5,000-customer sample were: k=3: 47,106, k=4: 38,181, k=5: 31,303, k=6: 28,999, k=7: 27,447. Silhouette was not used in the final report because this dataset is dominated by one-time customers, making business interpretability more reliable than a purely geometric score.

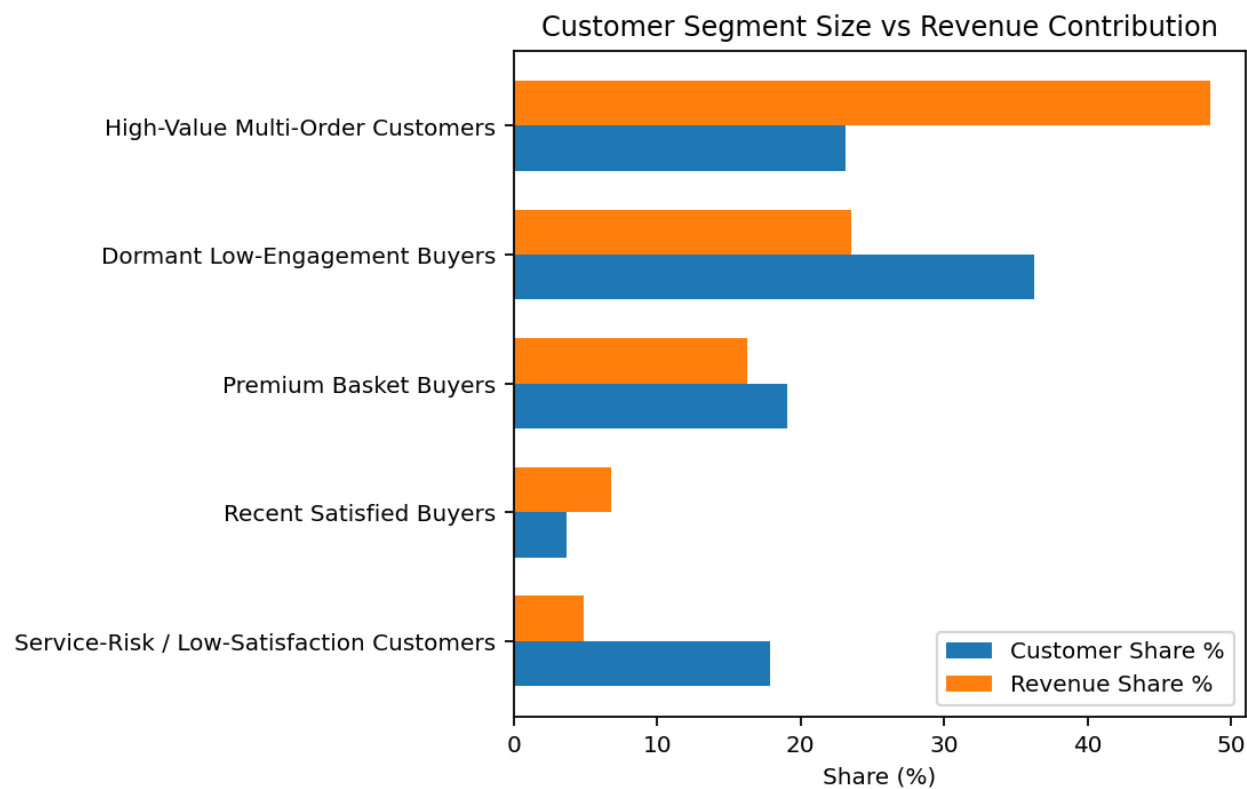


Table 2. Customer Segment Profile

Segment	Customers	Customer Share Pct	Total Revenue	Revenue Share Pct	Recency Days	Frequency	Monetary	Avg Order Value	Avg Review Score	Late Delivery Rate
High-Value Multi-Order Customers	21,594	23.1	7,489,044.85	48.6	234.00	1.00	346.81	346.81	3.68	0.18
Dormant Low-Engagement Buyers	33,828	36.2	3,626,115.10	23.5	204.00	1.00	107.19	107.19	4.53	0.01
Premium Basket Buyers	17,769	19.0	2,510,091.04	16.3	226.00	1.00	141.26	141.26	4.21	0.08
Recent Satisfied Buyers	3,480	3.7	1,043,076.51	6.8	198.00	1.90	299.73	168.76	3.97	0.06
Service-Risk / Low-Satisfaction Customers	16,687	17.9	754,277.73	4.9	224.00	1.00	45.20	45.20	4.00	0.12

Customer Segment Interpretation and Actions

High-Value Multi-Order Customers

Size: 21,594 customers (23.1% of customers) | Revenue share: 48.6% | Avg monetary value: 346.81 | Avg review: 3.68/5.

- Prioritize loyalty offers, early access to high-performing categories, and personalized cross-sell campaigns.
- Protect service quality for this group because small retention improvements have disproportionate revenue impact.

Dormant Low-Engagement Buyers

Size: 33,828 customers (36.2% of customers) | Revenue share: 23.5% | Avg monetary value: 107.19 | Avg review: 4.53/5.

- Use low-cost reactivation campaigns rather than heavy discounting.
- Suppress from frequent paid campaigns if response remains low to protect marketing ROI.

Premium Basket Buyers

Size: 17,769 customers (19.0% of customers) | Revenue share: 16.3% | Avg monetary value: 141.26 | Avg review: 4.21/5.

- Promote bundles, warranty/service add-ons, and premium product recommendations.
- Monitor freight costs because higher basket value can hide logistics inefficiency.

Recent Satisfied Buyers

Size: 3,480 customers (3.7% of customers) | Revenue share: 6.8% | Avg monetary value: 299.73 | Avg review: 3.97/5.

- Trigger second-purchase campaigns within 15-45 days of delivery.
- Recommend adjacent categories based on first purchase category and review positivity.

Service-Risk / Low-Satisfaction Customers

Size: 16,687 customers (17.9% of customers) | Revenue share: 4.9% | Avg monetary value: 45.20 | Avg review: 4.00/5.

- Run service recovery actions: apology coupon, faster support, proactive delivery updates.
- Investigate categories/sellers linked to low review scores and late delivery.

Table 3. Top States Within Each Customer Segment

Segment	State	Customers	Monetary
Dormant Low-Engagement Buyers	SP	17,176	1,747,900.96
Dormant Low-Engagement Buyers	MG	4,080	450,704.75
Dormant Low-Engagement Buyers	RJ	3,980	446,651.29
High-Value Multi-Order Customers	SP	6,769	2,392,528.63
High-Value Multi-Order Customers	RJ	3,259	1,062,567.82
High-Value Multi-Order Customers	MG	2,532	865,780.63
Premium Basket Buyers	SP	7,799	1,006,551.33
Premium Basket Buyers	MG	2,045	296,525.21

Premium Basket Buyers	RJ	1,987	285,066.50
Recent Satisfied Buyers	SP	1,500	410,909.34
Recent Satisfied Buyers	RJ	494	157,706.41
Recent Satisfied Buyers	MG	411	119,816.96
Service-Risk / Low-Satisfaction Customers	SP	5,906	214,113.33
Service-Risk / Low-Satisfaction Customers	RJ	2,193	103,523.16
Service-Risk / Low-Satisfaction Customers	MG	1,928	86,397.64

4. Product and Category Analysis

Product analysis was built from delivered order items and enriched with product master data, seller data, delivery outcomes, and review scores. The analysis uses product revenue, units sold, orders, sellers, freight burden, average price, review score, delivery speed, dimensions, and content attributes.

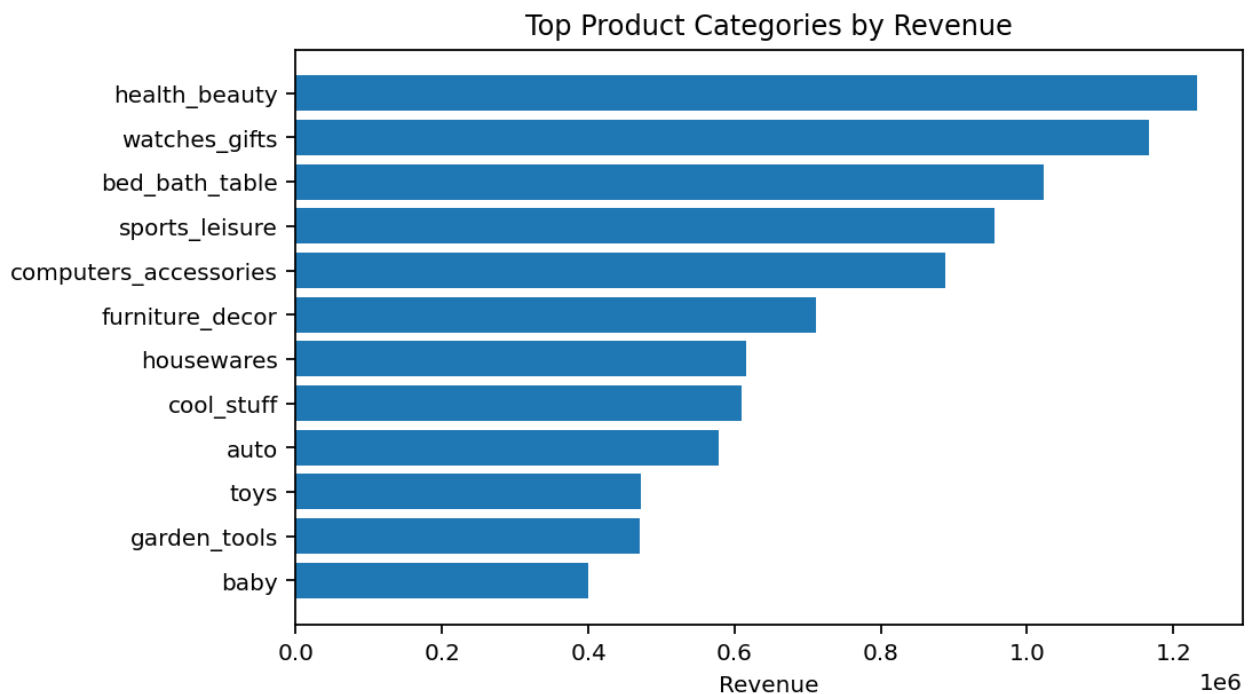
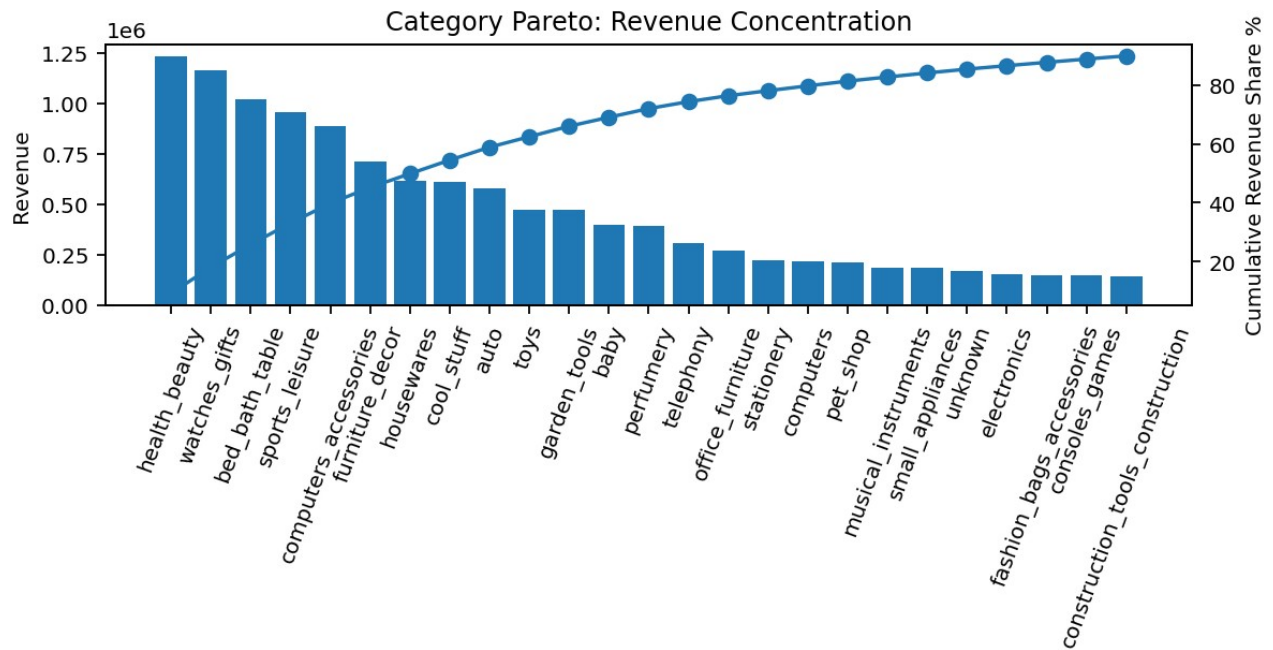


Table 4. Top Categories by Revenue

Category	Units Sold	Orders	Products	Sellers	Revenue	Revenue Share Pct	Avg Price	Avg Review Score	Late Delivery Rate
health_beauty	9,465	8,647	2,397	479	1,233,131.72	9.3	130.28	4.19	0.09
watches_gifts	5,859	5,495	1,300	95	1,166,176.98	8.8	199.04	4.07	0.08
bed_bath_table	10,953	9,272	2,991	189	1,023,434.76	7.7	93.44	3.92	0.08
sports_leisure	8,431	7,530	2,822	466	954,852.55	7.2	113.25	4.17	0.07
computers_accessories	7,644	6,530	1,600	279	888,724.61	6.7	116.26	3.99	0.08
furniture_	8,160	6,307	2,593	352	711,927.69	5.4	87.25	3.95	0.08

decor									
housewar es	6,795	5,743	2,282	452	615,628.69	4.7	90.60	4.11	0.06
cool_stuff	3,718	3,559	770	259	610,204.10	4.6	164.12	4.19	0.07
auto	4,140	3,810	1,853	371	578,966.65	4.4	139.85	4.12	0.08
toys	4,030	3,804	1,373	242	471,286.48	3.6	116.94	4.21	0.07
garden_to ols	4,268	3,448	725	227	470,495.28	3.6	110.24	4.08	0.08
baby	2,982	2,809	890	240	400,421.84	3.0	134.28	4.08	0.09



Product Segmentation

Products were segmented using business rules based on revenue, units sold, average price, freight-to-revenue ratio, and review score. This makes the segmentation easy to explain to non-technical stakeholders and directly usable for merchandising decisions.

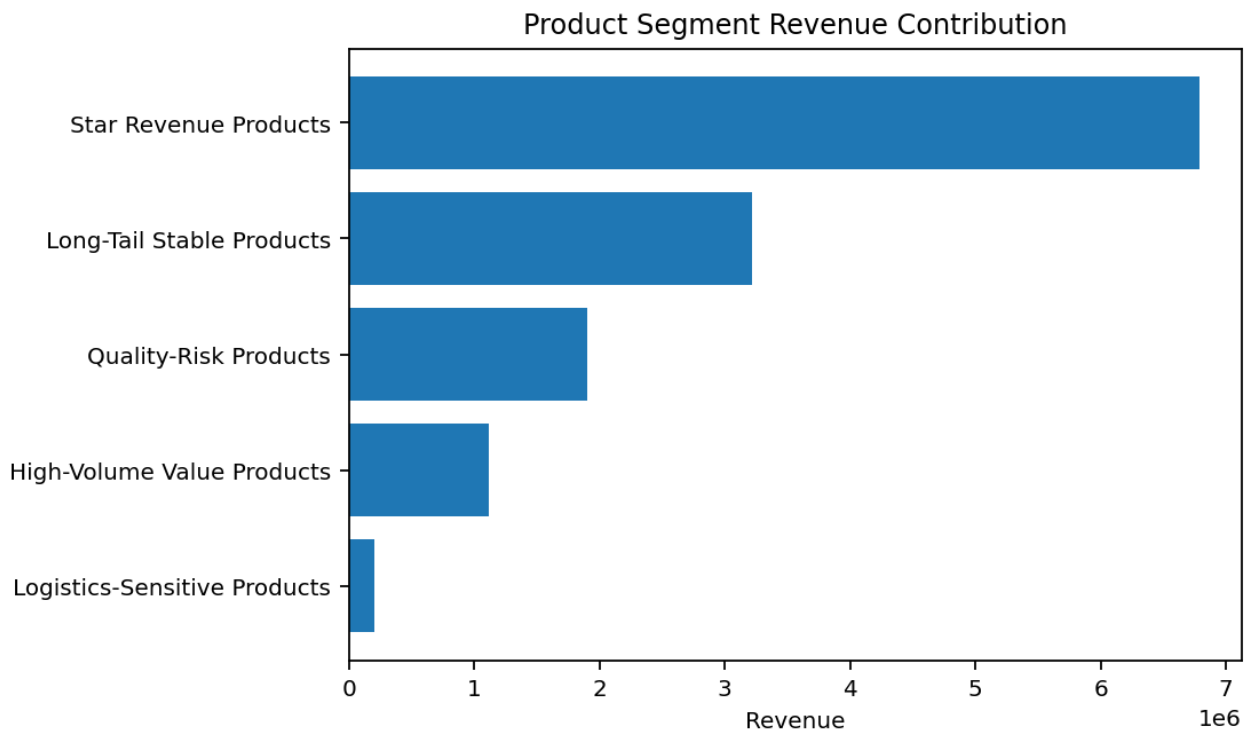


Table 5. Product Segment Summary

Product Segment	Products	Units Sold	Revenue	Avg Price	Avg Review Score	Freight To Revenue Ratio
Star Revenue Products	4,472	40,733	6,784,618.08	391.45	4.58	0.14
Long-Tail Stable Products	14,348	25,192	3,216,210.09	118.84	4.67	0.20
Quality-Risk Products	4,708	10,095	1,903,162.92	187.95	2.13	0.19
High-Volume Value Products	4,127	28,311	1,115,319.50	39.12	4.02	0.51
Logistics-Sensitive Products	4,561	5,866	202,187.52	32.82	4.07	0.81

Top Product-Level Opportunities

Table 6. Top Products by Revenue

Product Id	Category	Product Segment	Units Sold	Orders	Revenue	Avg Price	Avg Review Score	Freight To Revenue Ratio
bb50f2e236e5eea0100680137654686c	health_beauty	Star Revenue Products	194	186	63,560.00	327.63	4.24	0.06
6cdd53843498f92890544667809f1595	health_beauty	Star Revenue Products	153	148	53,652.30	350.67	4.36	0.08
d6160fb7873f184099d9bc95e30376af	computers	Star Revenue Products	33	33	45,949.35	1,392.40	4.67	0.03

d1c427060a0f73f6b889a5c7c61f2ac4	computers_accessories	Star Revenue Products	332	313	45,620.56	137.41	4.28	0.29
99a4788cb24856965c36a24e339b6058	bed_bath_table	Long-Tail Stable Products	477	456	42,049.66	88.15	3.93	0.19
3dd2a17168ec895c781a9191c1e95ad7	computers_accessories	Star Revenue Products	272	253	40,782.80	149.94	4.22	0.17
25c38557cf793876c5abd5931f922db	baby	Quality-Risk Products	38	38	38,907.32	1,023.88	2.68	0.04
5f504b3a1c75b73d6151be81eb05bdc9	cool_stuff	Star Revenue Products	63	63	37,733.90	598.95	4.56	0.11
53b36df67ebb7c41585e8d54d6772e08	watches_gifts	Star Revenue Products	321	304	37,454.63	116.68	4.20	0.06
aca2eb7d00ea1a7b8ebd4e68314663af	furniture_decor	Star Revenue Products	520	425	37,104.30	71.35	4.05	0.19
e0d64dcfaa3b6db5c54ca298ae101d05	watches_gifts	Long-Tail Stable Products	193	193	31,621.82	163.84	3.80	0.11
d285360f29ac7fd97640bf0baef03de0	watches_gifts	Long-Tail Stable Products	119	118	30,573.91	256.92	3.76	0.06
7a10781637204d8d10485c71a6108a2e	watches_gifts	Star Revenue Products	141	140	30,044.50	213.08	4.11	0.08
f1c7f353075ce59d8a6f3cf58f419c9c	bed_bath_table	Star Revenue Products	153	149	29,792.36	194.72	4.38	0.11
f819f0c84a64f02d3a5606ca95edd272	watches_gifts	Star Revenue Products	44	44	28,451.58	646.63	4.16	0.05

- health_beauty: strong revenue category with 9.3% share; maintain availability, improve search visibility, and monitor review/late delivery patterns.
- watches_gifts: strong revenue category with 8.8% share; maintain availability, improve search visibility, and monitor review/late delivery patterns.
- bed_bath_table: strong revenue category with 7.7% share; maintain availability, improve search visibility, and monitor review/late delivery patterns.
- sports_leisure: strong revenue category with 7.2% share; maintain availability, improve search visibility, and monitor review/late delivery patterns.
- computers_accessories: strong revenue category with 6.7% share; maintain availability, improve search visibility, and monitor review/late delivery patterns.

5. Geography, Payment and Operational Insights

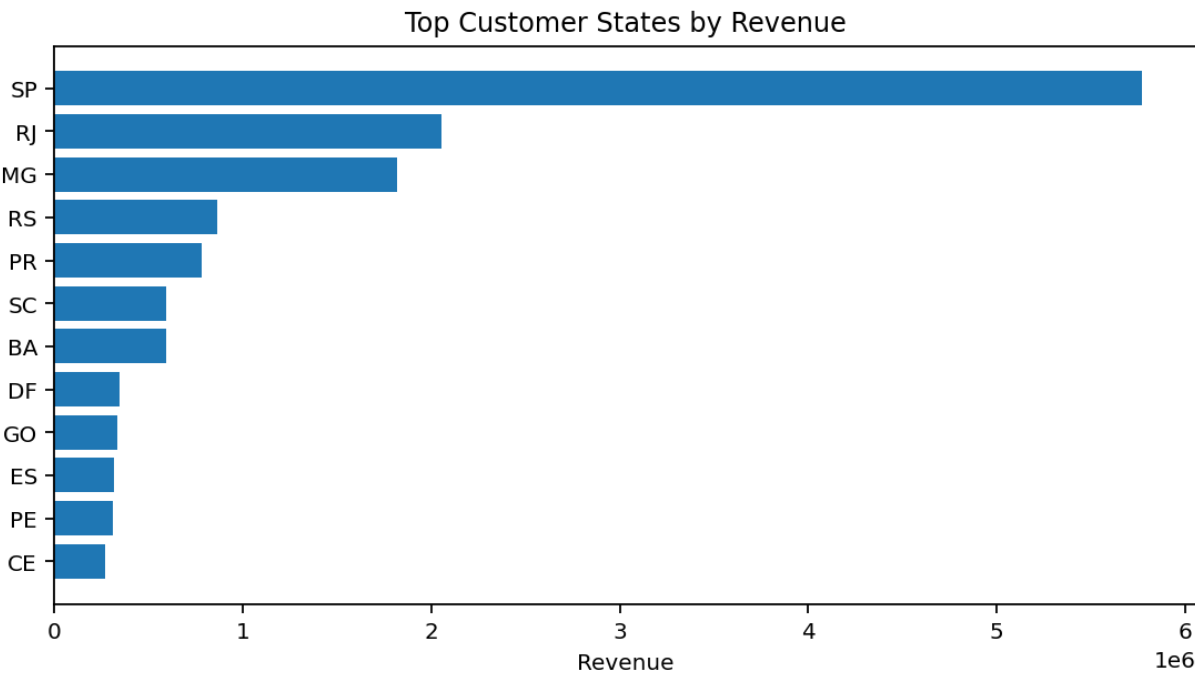


Table 7. Customer State Summary

Customer State	Customers	Orders	Revenue	Revenue Share Pct	Avg Order Value	Avg Review Score	Late Delivery Rate
SP	39,156	40,501	5,770,409.65	37.4	142.48	4.25	0.06
RJ	11,917	12,350	2,055,690.45	13.3	166.45	3.97	0.13
MG	11,001	11,354	1,819,277.61	11.8	160.23	4.19	0.06
RS	5,168	5,345	861,802.40	5.6	161.24	4.19	0.07
PR	4,769	4,923	781,919.55	5.1	158.83	4.24	0.05
SC	3,449	3,546	595,208.40	3.9	167.85	4.13	0.10
BA	3,158	3,256	591,270.60	3.8	181.59	3.93	0.14
DF	2,019	2,080	346,146.17	2.2	166.42	4.13	0.07
GO	1,895	1,957	334,294.22	2.2	170.82	4.10	0.08
ES	1,928	1,995	317,682.65	2.1	159.24	4.08	0.12
PE	1,551	1,593	309,074.59	2.0	194.02	4.08	0.11
CE	1,258	1,279	266,463.97	1.7	208.34	3.94	0.15

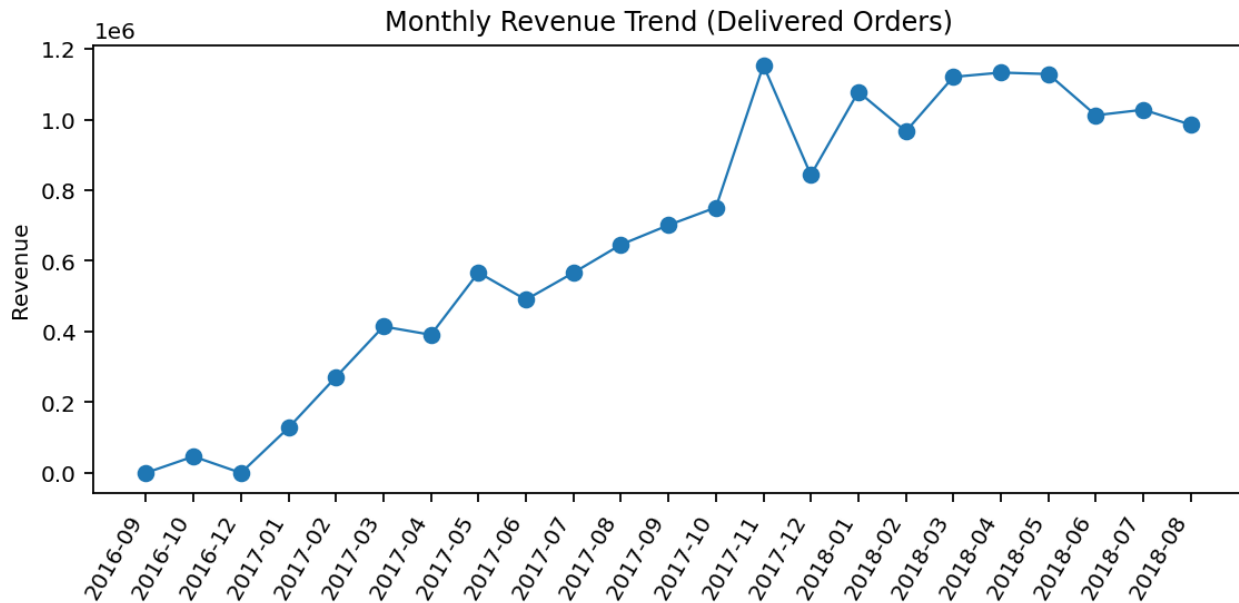
Table 8. Payment Method Summary

Payment Type	Transactions	Payment Value	Payment Share Pct
credit_card	76,795	12,542,084.19	78.3
boleto	19,784	2,869,361.27	17.9
voucher	5,775	379,436.87	2.4
debit_card	1,529	217,989.79	1.4
not_defined	3	0.00	0.0

Table 9. Order Status Distribution

Order Status	Orders	Share Pct
delivered	96,478	97.0
shipped	1,107	1.1
canceled	625	0.6

unavailable	609	0.6
invoiced	314	0.3
processing	301	0.3
created	5	0.0
approved	2	0.0



6. Business Recommendations

Table 10. Prioritized Recommendations

Area	Recommendation
Customer Retention	Create separate lifecycle campaigns for high-value multi-order customers, recent satisfied buyers, and dormant buyers. Avoid one generic campaign for all customers.
Service Recovery	Build a service-risk workflow for customers with low review scores and late delivery. Prioritize cases linked with high-value customers and high-revenue categories.
Merchandising	Protect stock depth for star revenue products and leading categories; use high-volume value products for acquisition and basket-building promotions.
Pricing and Freight	Monitor products with high freight-to-revenue ratios. These products may need shipping-fee optimization, regional seller matching, or bundle thresholds.
Catalog Governance	Improve missing product category/content/dimension completeness because these fields support search quality, recommendation performance, and logistics estimation.
Regional Growth	Use SP as the benchmark region, then identify underpenetrated high-population states with acceptable delivery performance for expansion campaigns.
Seller Management	For categories with low reviews or late deliveries, review seller-level SLA, product packaging, and carrier performance before scaling paid promotion.

7. Limitations and Next Steps

- The dataset ends in 2018, so findings are historical and should be used for internship/project analytics rather than current market decisions.
- Customer demographic attributes such as age, gender, income, and marketing campaign exposure are not available; clustering is therefore behavior-based, not demographic.
- Profit or margin data is not provided; revenue-driving products may not be the most profitable products.
- Geolocation was aggregated by zip-code-prefix, which is appropriate for regional analysis but not exact customer-location analysis.
- Future work can add CLTV modeling, churn prediction, cohort retention, market-basket analysis, and seller scorecards using this cleaned analytical base.

Appendix A. Analytical Data Dictionary

Output File	Description
customer_clustering_output.csv	One row per customer_unique_id with RFM, review, delivery, payment, freight and cluster label features.
customer_cluster_profile.csv	Segment-level summary table used for interpretation and recommendations.
product_analysis_output.csv	One row per product_id with sales, reviews, freight, logistics and product segment fields.
category_summary.csv	Category-level unit, order, seller, product, revenue and service metrics.
cleaned_orders_enriched.csv	Order-level enriched dataset joining customers, orders, items, payments, reviews and geolocation aggregates.
state_summary.csv	Customer-state level revenue, orders, customers and service metrics.
monthly_summary.csv	Monthly orders, revenue and customer counts for delivered orders.